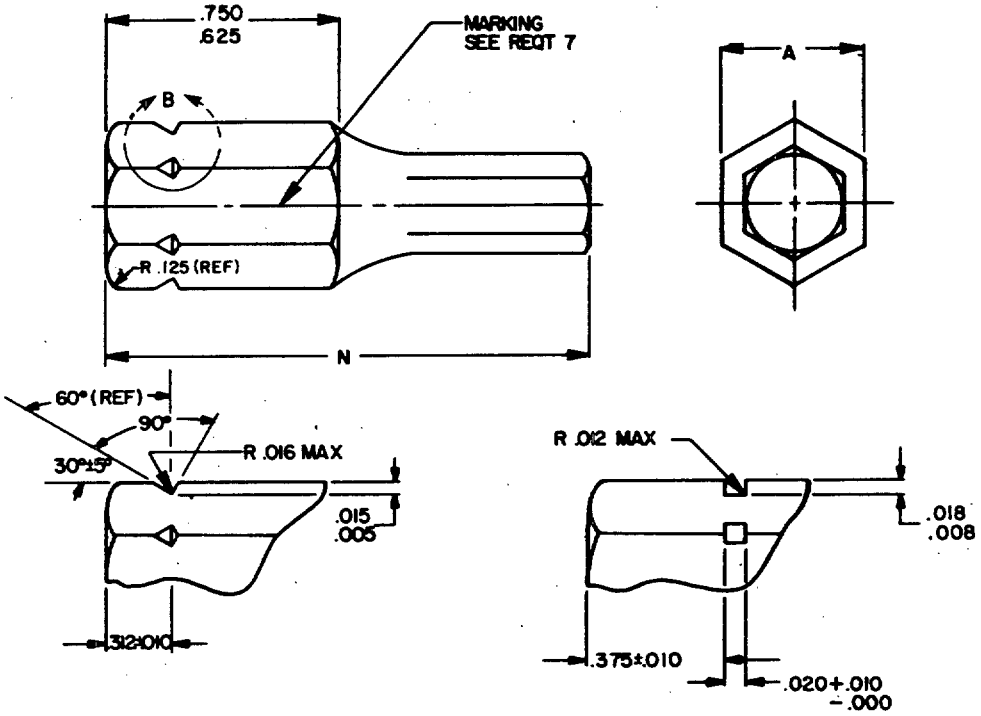
MILITARY SPECIFICATION SHEET

BIT, SCREWDRIVER, HEX, EXTERNAL HEX SHANK, MANUALLY DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the bits described herein shall consist of this specification and the latest issue of MIL-B-9946.



VIEW B-RETAINING GROOVE-NOTCH TYP SIX PLACES

NOTES:
1. All dimensions are in inches
2. Bit, retaining groove and tip point shall be in accordance with TABLE I.

FIGURE 1. BIT, SCREWDRIVER, HEX, EXTERNAL HEX SHANK, MANUALLY DRIVEN

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TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	BIT DIMENSIONS								TIP DIMENSIONS						TORSIONAL MOMENT LBF - IN
	DRIVER SIZE MAX	RECESS SIZE NOM	SCREW DIAMETER SIZE NOMINAL REF				A HEX +.000 -.004	N LENGTH ±.120	W HEX WIDTH ACROSS FLATS		Y HEX WIDTH ACROSS CORNERS		L HEX LENGTH SCREW DRIVING SURFACE	K CHAMFER	
			CAP SCREW	FLAT CTSK HEAD	BUTTON HEAD	SET SCREW			MAX	MIN	MAX	MIN	MIN	MAX	
062	.062	.062	#1	#3 & #4	#3 & #4	#5 & #6	.250	1.000	.0625	.0615	.0701	.0680	.187	.008	10.5
078	.078	.078	#2 & #3	#5 & #6	#5 & #6	#8	.250	1.000	.0781	.0771	.0880	.0859	.187	.008	21.0
094	.094	.094	#4 & #5	#8	#8	#10	.250	1.000	.0937	.0927	.1058	.1035	.218	.009	37.0
109	.109	.109	#6	-	-	-	.250	1.000	.1094	.1079	.1238	.1210	.250	.014	59.0
125	.125	.125	#8	#10	#10	1/4	.250	1.000	.1250	.1235	.1418	.1390	.250	.015	86.0
141	.141	.141	#8	-	-	-	.250	1.000	.1406	.1391	.1593	.1566	.250	.016	125.0
156	.156	.156	#10	1/4	1/4	5/16	.250	1.312	.1562	.1547	.1774	.1745	.250	.016	170.0
156A	.156	.156	#10	1/4	1/4	5/16	.312	1.312	.1562	.1547	.1774	.1745	.250	.022	170.0
188	.188	.188	1/4	5/16	5/16	3/8	.250	1.312	.1875	.1860	.2135	.2105	.281	.022	295.0
188A	.188	.188	1/4	5/16	5/16	3/8	.312	1.312	.1875	.1860	.2135	.2105	.281	.022	295.0
219	.219	.219	-	3/8	3/8	7/16	.250	1.312	.2187	.2172	.2490	.2460	.343	.024	425.0
219A	.219	.219	-	3/8	3/8	7/16	.312	1.312	.2187	.2172	.2490	.2460	.343	.024	425.0
250	.250	.250	5/16	7/16	-	1/2&9/16	.250	1.312	.2500	.2485	.2845	.2815	.343	.030	630.0
250A	.250	.250	5/16	7/16	-	1/2&9/16	.312	1.312	.2500	.2485	.2845	.2815	.343	.030	630.0
312 1/	.312	.312	3/8	1/2	1/2	5/8	.250	1.312	.3125	.3110	.3570	.3531	.406	.032	1,250.0
312A	.312	.312	3/8	1/2	1/2	5/8	.312	1.312	.3125	.3110	.3570	.3531	.406	.032	1,250.0

1/ TOOL BIT POINT IS LARGER THAN SHANK DRIVE

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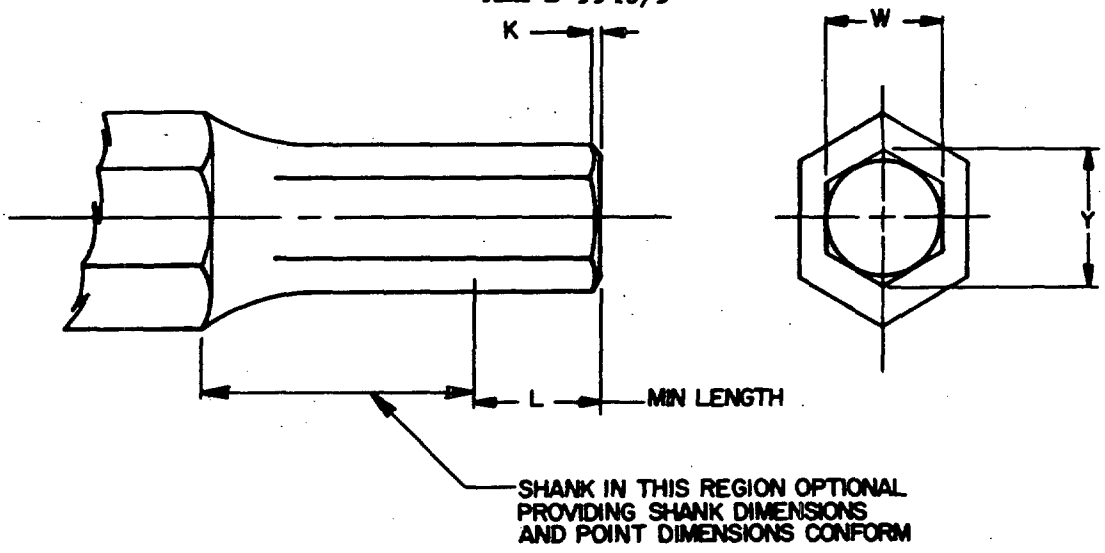


FIGURE 2. TIP POINT

REQUIREMENTS:

1. Material: Alloy Tool Steel in accordance with American Society for Testing Material (ASTM) A681, Shock-Resisting Steel, Type S-2, (UNS T41902).

2. Hardness: Macrohardness, (Rockwell C Scale) 56-62HRC. Three macrohardness readings shall be taken at the location shown in FIGURE 3. Each hardness value must be within the range specified.

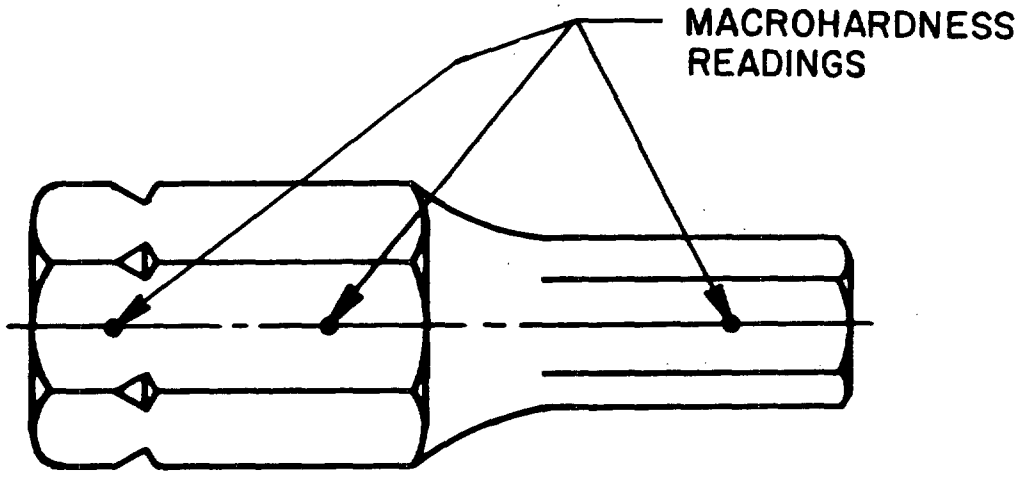


FIGURE 3. MACROHARDNESS READING LOCATIONS

Superficial hardness, (Rockwell HR15N scale) 87-90. Superficial hardness readings shall be taken on bid samples at two locations on sectionalized specimens along the blade.

3. Driver bit gage: Driver bit gage shall be in accordance with FIGURE 4 and TABLE II.

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4. Tip point: Tip point shall be in accordance with FIGURE 2 and TABLE I.

5. Test block: Test block shall be in accordance with FIGURE 5 and TABLE III. Torque tests are conducted using this block as described herein and in MIL-B-9946.

6. Torsion test: When tested, all bits shall withstand the test load specified in TABLE I without injury or permanent deformation which may affect their durability or serviceability. A torsion test shall be conducted to determine conformance with the test load specified in TABLE I. The bit shall be inserted into a female socket designed for the size bit being tested. The bit shall withstand the torsional moment (test load) specified in TABLE I. An angular reference line shall be established and the bits shall show less than a 5° permament angular deflection. The bit shall show no flaws, cracks or other defects which might affect its durability or serviceability. The part of the bit inserted into the socket shall show no distortion or rounded corners. For bit sizes .062 inches thru .250 inches, inclusive, the torsional load shall be further increased until the bit fails. Failure shall occur by a uniform shear fracture for these sizes. Sample bits shall be tested using a test recess conforming to FIGURE 5 and TABLE III. There shall be no debris present in the recess during the test.

7. Identification marking: Driver bits shall be marked by etching, stamping or engraving onto each bit the following minimum information:

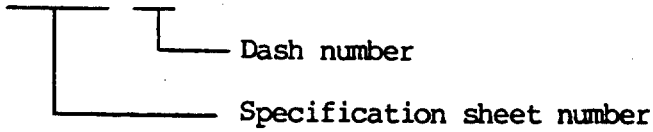
   a. Military specification part number.

   b. Manufacturer's symbol or code.

   c. Country of manufacture.

8. Part number: The part number consists of the letter "M", the basic number of this specification sheet, followed by a dash number taken from TABLE I.

   Example: M9946/9-125

   M9946/9-125 indicates - Bit, Screwdriver, Hex, External Hex Shank, Manually Driven; Driver Size Max .125; Recess Size Nom .125; Screw Diameter Size Nom Ref, #8, #10, #10, 1/4; Hex Size .250; Bit Length 1.000; Tip Width Across Flats .1250 Max - .1235 Min; Width Across Corners .1418 Max - .1390 Min; Length Screw Driving Surface .250; Chamfer .015 Max.

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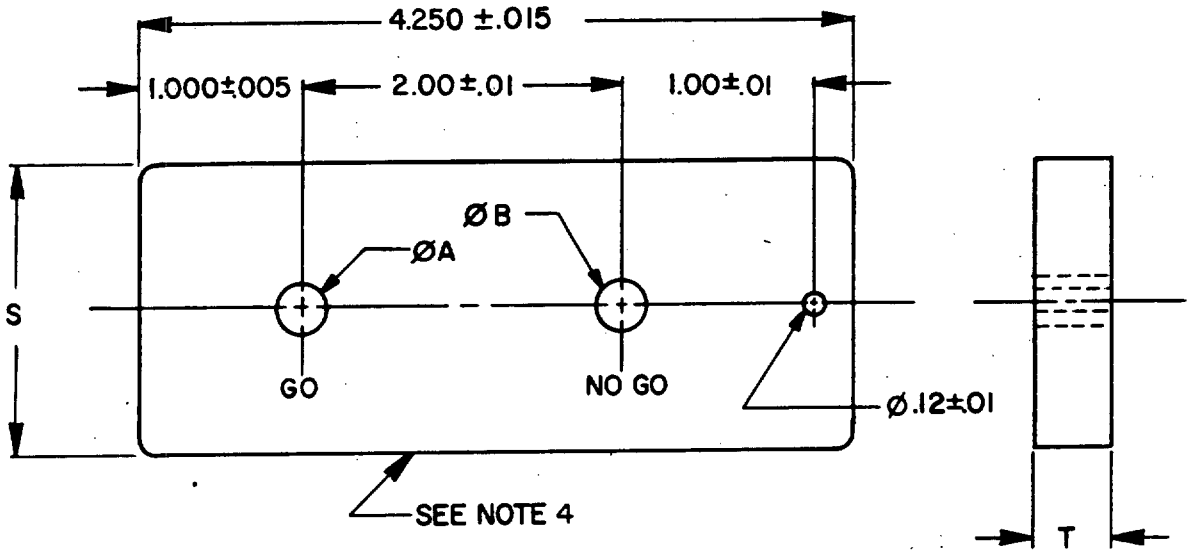


FIGURE 4. DRIVER BIT GAGE

TABLE II. DRIVER BIT GAGE DIMENSIONS FOR HEX BITS

KEY AND SOCKET SIZE NOMINAL	GO ØA MAX	GO ØA MIN	NO GO ØB MAX	NO GO ØB MIN	S MIN	T ±.005
.062	.0704	.0701	.0680	.0677	1.00	.187
.078	.0883	.0880	.0859	.0856	1.00	.187
.094	.1061	.1058	.1035	.1032	1.00	.187
.109	.1241	.1238	.1210	.1207	1.00	.187
.125	.1421	.1418	.1390	.1387	1.00	.187
.141	.1596	.1593	.1566	.1563	1.00	.187
.156	.1777	.1774	.1745	.1742	1.00	.375
.188	.2138	.2135	.2105	.2102	1.00	.375
.219	.2493	.2490	.2460	.2457	1.00	.375
.250	.2848	.2845	.2815	.2812	1.00	.375
.312	.3570	.3570	.3531	.3528	1.00	.375

NOTES:
1. All dimensions are in inches.
2. Material: CRES in accordance with ASTM A276, Type 440C (UNS S44004).
3. Hardness: 58-62HRC
4. Marking: Stamp nominal key and socket size and "GO" and "NO GO".

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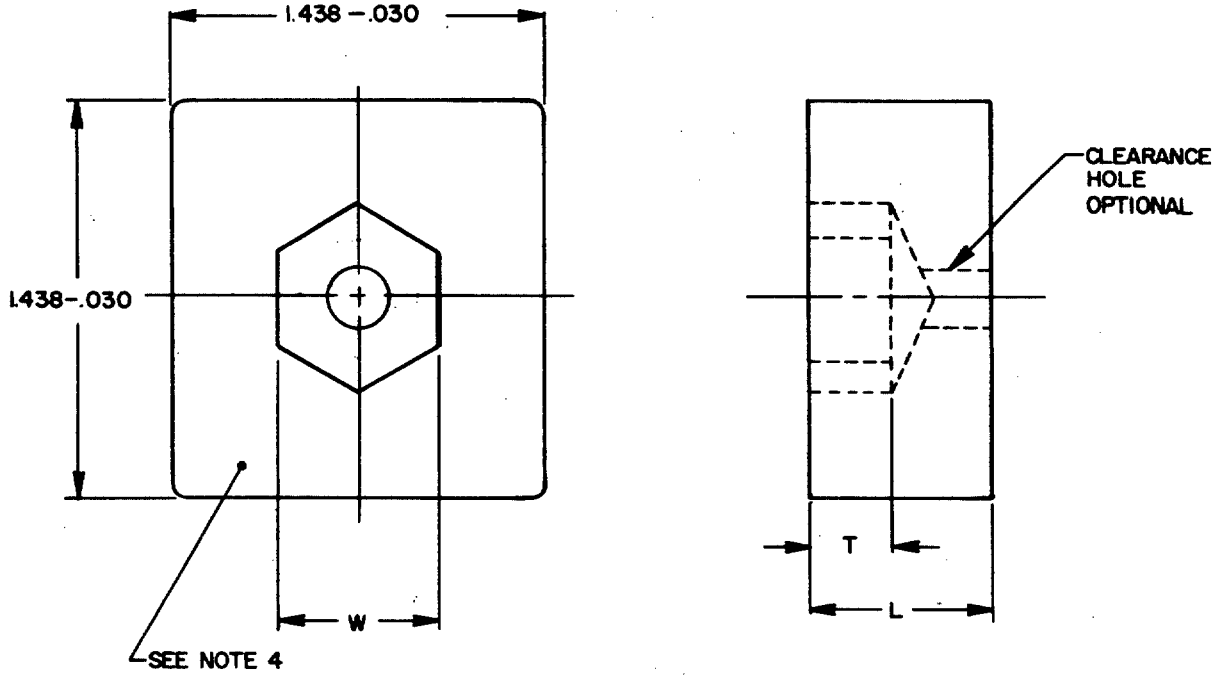


FIGURE 5. TEST BLOCK

TABLE III. TEST BLOCK DIMENSIONS FOR HEX BITS

KEY AND SOCKET SIZE NOMINAL	W HEX WIDTH ACROSS FLATS		T MINIMUM ENGAGEMENT STRAIGHT SIDES	L THICKNESS REF
	MAX	MIN		
.062	.0635	.0625	.062	.187
.078	.0791	.0781	.078	.187
.094	.0952	.0937	.094	.187
.109	.1111	.1094	.109	.187
.125	.1270	.1250	.125	.375
.141	.1426	.1406	.141	.375
.156	.1587	.1562	.156	.375
.188	.1900	.1875	.188	.375
.219	.2217	.2187	.219	.375
.250	.2530	.2500	.250	.375
.312	.3160	.3125	.312	.750

NOTES:
1. All dimensions are in inches.
2. Material: Alloy Tool Steel in accordance with ASTM A681, Type S-2 (UNS T41902).
3. Hardness: 62-64HRC
4. Marking: Stamp nominal key and socket size.

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